AOI-Inspection — Standard Operating Procedure

AOI-Inspection — Standard Operating Procedure
1. Purpose
2. Normal operating envelope
3. Idle reasons
4. Faults & corrective actions
5. Preventive maintenance schedule
6. References

AOI-Inspection — Standard Operating Procedure

> **Audience:** Quality Worker · Maintenance Tech · Line Manager
> **Applies to:** Line-E, station 4, AOI-Inspection (Automated Optical Inspection)

1. Purpose

AOI-Inspection performs automated optical inspection of the soldered SMT joints leaving Reflow-Oven. Reject disposition follows `Quality_Reject_Disposition_Policy.pdf`.

2. Normal operating envelope

- Ideal cycle time: **10 s**
- Max acceptable cycle time: **14 s**
- Expected reject rate: **≤ 4 %**
- Fault probability per cycle: **0.3 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Reflow-Oven (station 3) is not feeding boards.	Verify Reflow-Oven state.
`Blocked`	Through-Hole-Insert (station 5) buffer full.	Verify Through-Hole-Insert state.

4. Faults & corrective actions

4.1 `Camera-Fault`

- **Symptoms in telemetry:** `fault_type = "Camera-Fault"`. Inspection cycle aborts or image acquisition fails.
- **Likely root cause:** lens fouled, focus drifted, camera failure.
- **Corrective action:**
  1. Clean lens; cycle camera power.

2. Replace camera if acquisition still fails.
- **Restart criteria:** master PCB passes recipe.

4.2 `Light-Fail`

- **Symptoms in telemetry:** `fault_type = "Light-Fail"`. Image contrast drops.
- **Likely root cause:** ring-light LED bank failed.
- **Corrective action:**
  1. LOTO. Replace the light bar.
  2. Restore brightness setpoint.
- **Restart criteria:** image meets contrast threshold.

4.3 `Calibration-Drift`

- **Symptoms in telemetry:** `fault_type = "Calibration-Drift"`. False-positive or false-negative rejects rise.
- **Likely root cause:** classifier drift; temperature-induced offset.
- **Corrective action:**
  1. Run the calibration set (known-good + known-defect PCBs).
  2. Tune thresholds; sign off recipe revision in the QMS.
- **Restart criteria:** 100 % accuracy on the calibration set.

5. Preventive maintenance schedule

- **Daily:** master PCB verification at shift start.
- **Weekly:** clean lenses and diffusers.
- **Monthly:** classifier retraining.
- **Quarterly:** lighting replacement.

6. References

- `Quality_Reject_Disposition_Policy.pdf`
- `Shift_Handover_Checklist.pdf`
- `Line-E_03_Reflow-Oven_SOP.pdf`
- `Line-E_05_Through-Hole-Insert_SOP.pdf`